

MOÏSE BLANCHARD

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ACADEMIC HISTORY

Georgia Institute of Technology, Atlanta, GA 2025 – to date
TENNENBAUM EARLY CAREER PROFESSOR AND ASSISTANT PROFESSOR
H. Milton Stewart School of Industrial and Systems Engineering (ISyE)

Columbia University, New York, NY 2024 – 2025
POSTDOCTORAL FELLOWS PROGRAM
Data Science Institute (DSI)

Massachusetts Institute of Technology, Cambridge, MA 2019 – 2024
PH.D. STUDENT IN OPERATIONS RESEARCH
Operations Research Center (ORC), Laboratory for Information & Decision Systems (LIDS). GPA 5.0/5.0
Advisor: Prof. Patrick Jaillet

École Polytechnique, Palaiseau, France 2016 – 2020
B.SC. AND M.SC. IN APPLIED MATHEMATICS, **Valedictorian**
Minor: Computer Science, Mathematics, Physics. GPA 4.0/4.0

Lycée Louis-le-Grand, Paris, France 2014 – 2016
CLASSES PRÉPARATOIRES
Mathematics, Physics, and Computer Science. GPA 4.0/4.0

RESEARCH INTERESTS

STATISTICAL DECISION-MAKING: online learning, contextual bandits

MACHINE LEARNING: learning with unstructured data, algorithm design, high-dimensional statistics

OPTIMIZATION: optimization under memory constraints, discrete optimization

PUBLICATIONS (authorship order by contribution, equal contribution marked with *)

JOURNAL PUBLICATIONS

- J1 Moïse Blanchard, Patrick Jaillet. Universal regression with adversarial responses.
Annals of Statistics (AoS), 2023
- J2 Moïse Blanchard*, Adam Q. Jaffe*. Fréchet mean set estimation in the Hausdorff metric, via relaxation.
Bernoulli, 2023
- J3 Moïse Blanchard, Junhui Zhang, Patrick Jaillet. Quadratic memory is necessary for optimal query complexity in convex optimization: Center-of-mass is Pareto-optimal.
Mathematics of Operations Research (MOR), 2023
- J4 Moïse Blanchard, Alexandre Jacquillat, Patrick Jaillet. Probabilistic bounds on the k-Traveling Salesman Problem and the Traveling Repairman Problem.
Mathematics of Operations Research (MOR), 2022
Winner of INFORMS Transportation Science & Logistics (TSL) 2023 Best student paper award
- J5 Moïse Blanchard, Jesús A. De Loera, Quentin Louveaux. On the length of monotone paths in polyhedra.
SIAM Journal on Discrete Mathematics, 2021
Winner of the Rivot medal for outstanding research, French Science Academia

CONFERENCE PUBLICATIONS

- C1 Moïse Blanchard. Agnostic smoothed online learning.
Symposium on Theory of Computing (STOC), 2025
- C2 Yan Dai*, Moïse Blanchard*, Patrick Jaillet. Non-Monetary Mechanism Design without Distributional Information: Using Scarce Audits Wisely.
Conference on Learning Theory (COLT), 2025
- C3 Noah Krever, Jakub Černý, Moïse Blanchard, Christian Kroer. GUARD: Constructing Realistic Two-Player Matrix and Security Games for Benchmarking Game-Theoretic Algorithms.
The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS), 2025
Spotlight at NeurIPS in Datasets and Benchmark Track
- C4 Moïse Blanchard. Gradient Descent is Pareto-Optimal in the Oracle Complexity and Memory Tradeoff for Feasibility Problems.
Symposium on Foundations of Computer Science (FOCS), 2024
- C5 Moïse Blanchard*, Doron Cohen*, Aryeh Kontorovich*. Correlated Bernoulli Process.
Conference on Learning Theory (COLT), 2024
- C6 Moïse Blanchard, Václav Voráček. Tight bounds for local Gliveko-Cantelli.
International Conference on Algorithmic Learning Theory (ALT), 2024
- C7 Moïse Blanchard, Junhui Zhang, Patrick Jaillet. Quadratic memory is necessary for optimal query complexity in convex optimization: Center-of-mass is Pareto-optimal.
Conference on Learning Theory (COLT), 2023
- C8 Moïse Blanchard, Junhui Zhang, Patrick Jaillet. Memory-constrained algorithms for convex optimization.
Advances in Neural Information Processing Systems (NeurIPS), 2023
- C9 Moïse Blanchard. Universal online learning: An optimistically universal learning rule.
Conference on Learning Theory (COLT), 2022
Best Student Paper Runner-up Award, COLT
- C10 Moïse Blanchard*, Romain Cosson*. Universal online learning with bounded loss: Reduction to binary classification.
Conference on Learning Theory (COLT), 2022
- C11 Moïse Blanchard*, Romain Cosson*, Steve Hanneke. Universal online learning with unbounded losses: Memory is all you need.
International Conference on Algorithmic Learning Theory (ALT), 2022
- C12 Moïse Blanchard, Amine Bennouna. Shallow and deep networks are near-optimal approximators of Korobov functions.
International Conference on Learning Representations (ICLR), 2022

PREPRINTS

- P1 Moïse Blanchard, Vineet Goyal. Collaborative Min-Max Regret in Grouped Multi-Armed Bandits.
Submitted to Operations Research, 2025
- P2 Moïse Blanchard, Samory Kpotufe. Distributionally-Constrained Adversaries in Online Learning.
Submitted to Journal of Machine Learning Research (JMLR), 2025
- P3 Moïse Blanchard*, Adam Quinn Jaffe*, Nikita Zhivotovskiy. Consistency and Inconsistency in k -Means Clustering.
Submitted to Electronic Journal of Statistics, 2025
- P4 Moïse Blanchard, Patrick Jaillet. Near-optimal mechanisms for resource allocation without monetary transfers.
Major revision at Operations Research, 2024
- P5 Moïse Blanchard, Steve Hanneke, Patrick Jaillet. Adversarial Rewards in Universal Learning for Contextual Bandits.
Major revision at Journal of Machine Learning Research (JMLR), 2024
- P6 Moïse Blanchard, Steve Hanneke, Patrick Jaillet. Contextual bandits and optimistically universal learning.
Major revision at Mathematics of Operations Research, 2024

TALKS

AGNOSTIC SMOOTHED ONLINE LEARNING

- Symposium on Theory of Computing (STOC) *June 2025*

GRADIENT DESCENT IS PARETO-OPTIMAL IN THE ORACLE COMPLEXITY AND MEMORY TRADEOFF FOR FEASIBILITY PROBLEMS

- INFORMS Annual Meeting *Oct 2025*
- Invited talk at Workshop on Information and Learning (WIL), London Business School *June 2025*
- Invited talk at University of Southern California, Computer science department *June 2025*
- Invited talk at Georgia Institute of Technology, ISyE. Discrete Optimization Seminar *Mar 2025*
- Symposium on Foundations of Computer Science (FOCS) *Oct 2024*
- Invited Talk at Google. Learning Theory Seminar *Aug 2024*

NEAR-OPTIMAL MECHANISMS FOR RESOURCE ALLOCATION WITHOUT MONETARY TRANSFERS

- INFORMS Annual Meeting *Oct 2024*

UNIVERSAL LEARNING: CONSISTENCY UNDER MINIMAL DATA ASSUMPTIONS

- Seminar class for 1st year PhD students, Georgia Institute of Technology, ISyE *Sep 2025*
- Invited talk at Columbia University, IEOR. Applied Probability Seminar *Oct 2024*

CORRELATED BERNOULLI PROCESS

- Conference on Learning Theory (COLT) *Jul 2024*

TIGHT BOUNDS FOR GLIVENKO-CANTELLI

- International Conference on Algorithmic Learning Theory (ALT) *Feb 2024*

UNIVERSAL LEARNING FOR DECISION-MAKING

- Invited talk at Imperial College London, Analytics and Operations *Jan 2024*
- Invited talk at University of California San Diego, HDSI *Jan 2024*
- Invited talk at University of Southern California, ISE *Jan 2024*
- Invited talk at Duke University, Fuqua Business School, Decision Sciences *Jan 2024*
- Invited talk at New York University, Stern Business School, OM *Jan 2024*
- Invited talk at Georgia Institute of Technology, ISyE *Dec 2023*
- Invited talk at Northwestern University, Kellogg Business School, Operations *Nov 2023*

MEMORY-CONSTRAINED ALGORITHMS FOR CONVEX OPTIMIZATION

- Conference on Neural Information Processing Systems (NeurIPS) *Dec 2023*

QUADRATIC MEMORY IS NECESSARY FOR OPTIMAL QUERY COMPLEXITY IN CONVEX OPTIMIZATION: CENTER-OF-MASS IS PARETO-OPTIMAL

- Conference on Learning Theory (COLT) *Jul 2023*
- Invited talk at Toyota Technological Institute at Chicago (TTIC) *Apr 2023*
- LIDS & Statistics Student Seminar, MIT *Mar 2023*

CONTEXTUAL BANDITS AND OPTIMISTICALLY UNIVERSAL LEARNING

- INFORMS Annual Meeting *Oct 2023*
- Cornell Young Research Workshop, Cornell ORIE *Oct 2023*
- LIDS & Statistics Student Seminar, MIT *Nov 2022*
- Operations Research Student Seminar, MIT *Oct 2022*

UNIVERSAL REGRESSION WITH ADVERSARIAL RESPONSES

- INFORMS Annual Meeting *Oct 2022*
- Invited talk at Université Pierre et Marie Curie, Laboratoire Jacques-Louis Lions *Mar 2022*
- LIDS Conference, MIT *Feb 2022*

REALIZABLE ONLINE LEARNING WITH MINIMAL ASSUMPTIONS

- Conference on Learning Theory (COLT): “Universal online learning: an optimistically universal learning rule” *Jul 2022*
- Conference on Learning Theory (COLT): “Universal online learning with bounded loss: Reduction to binary classification” *Jul 2022*
- International Conference on Algorithmic Learning Theory (ALT) : “Universal online learning with unbounded loss: memory is all you need” *Mar 2022*
- Operations Research Student Seminar, MIT *Nov 2021*

PROBABILISTIC BOUNDS ON THE K-TRAVELING SALESMAN AND TRAVELING REPAIRMAN PROBLEMS

- INFORMS Annual Meeting, Transportation Science & Logistics award session *Oct 2023*

THE REPRESENTATION POWER OF NEURAL NETWORKS FOR KOROBV FUNCTIONS

- International Conference on Learning Representations (ICLR) *Apr 2021*
- MIT SIAM Conference *Oct 2021*
- Operations Research Student Seminar, MIT *Mar 2021*

ONLINE MATCHINGS ON UNKNOWN BIPARTITE GRAPHS

- INFORMS Annual Meeting *Oct 2020*

TEACHING EXPERIENCE

Instructor: Computational Statistics (ISyE 6416), *Georgia Tech* *Fall 2025*

- Qualifying course for PhD in Statistics (ISyE)

Teaching Assistant: Optimization Methods (6.215/15.093J), *MIT* *Fall 2021*

- Qualifying course for PhD in Electrical Engineering and Computer Science (EECS)
- Core course of the Master of Business Analytics (MBAn)

Teaching Assistant: Advanced Analytics Edge (MBAn) (15.072), *MIT* *Fall 2020*

- Core course of the Master of Business Analytics (MBAn)

Instructor for Classes Préparatoires, *Lycée Condorcet and Henry IV, Paris, France* *2017 - 2019*

- Led and evaluated oral exams for undergraduate students in mathematics
- Designed practice material in real and complex analysis, algebra, differential equations, and probability

GRANTS AND AWARDS

- Columbia University Data Science Institute (DSI) Postdoctoral fellowship *2024*
- Winner INFORMS Transportation Science & Logistics (TSL) Best student paper award *2023*
- **Air Force Office of Scientific Research Grant**, with Prof. Patrick Jaillet *2023*
- Prize for solving a COLT 2019 open problem in Memory-constrained Convex Optimization *2023*
- **COLT Best Student Paper Runner-Up Award** *2022*
- Prize for solving COLT 2021 open problems in Universal Learning *2022*
- Best Presentation Award, Laboratory of Information and Decision Systems Conference, MIT *2022*
- DeepMind Student Grant for COLT *2022*
- Bronze medal, Alibaba Global Mathematics Competition *2022*
- Honorable mention, Alibaba Global Mathematics Competition *2021*
- 2nd Prize, The East Coast Data Open by Citadel *2020*
- Laplace medal given to Valedictorian of École Polytechnique, French Science Academia *2019*
- Rivot medal for outstanding research at École Polytechnique, French Science Academia *2019*
- Bronze medal, International Physics Olympiad (IPhO) *2015*
- Bronze medal, International Mathematics Olympiad (IMO) *2014*
- Silver medal, Junior Balkanic Mathematics Olympiad (JBMO) *2014*
- 1st Prize, Concours Général in Mathematics, France *2014*

WORK AND RESEARCH EXPERIENCE

Georgia Institute of Technology, ISyE, Assistant Professor, Atlanta, GA *Aug 2025 – to date*

Columbia University, DSI, Postdoctoral research scientist, New York, NY *Aug 2024 – June 2025*

MIT, ORC & LIDS, Research assistant, Cambridge, MA *Sep 2019 – May 2024*

- Designed general-use algorithms for learning unrestricted tasks with minimal assumptions, for supervised learning and contextual bandits
- Analyzed the impacts of memory constraints for optimization and provided memory-efficient algorithms

Amazon Inc., Research intern, Cambridge, MA *Apr 2020 – Jul 2020*

- Co-developed a prediction model for the value of new vendors for Amazon’s vendor flex program
- Developed a mixed-integer optimization formulation to inform the selection of new vendors

University of California, Davis, Research assistant with Prof. Jesús De Loera, Davis, CA *Apr 2019 – Aug 2019*

- Analyzed conditions for the fast convergence of simplex methods in combinatorial problems by studying the length of monotone paths
- Supervised two undergraduate students to conduct empirical research on fast pivot rules for combinatorial polyhedra

INRIA, Undergraduate researcher with Prof. Laurent Massoulié, Paris, France *Sep 2018 – Mar 2019*

- Provided theoretical analysis for the problem of reconstructing lattice graphs from local neighborhoods, tightening phase transition bounds from previous works
- Developed algorithms for efficient lattice recovery

École Polytechnique, Undergraduate researcher with Prof. Gabriel Peyré, Palaiseau, France *Sep 2017 – Mar 2018*

- Applied optimal transport to natural language recognition and classification

REVIEWING SERVICE

JOURNALS: Operations Research, Mathematics of Operations Research, INFORMS Journal on Optimization, INFORMS Journal on Computing, Machine Learning, Journal of the ACM, Journal of Machine Learning Research, Discrete Optimization

CONFERENCES: COLT (2024-2025), SODA (2025), NeurIPS (2021-2025), ICML (2023-2024), ICLR (2022), ALT (2023-2024)

REFERENCES

PATRICK JAILLET, MIT

Professor of Electrical Engineering and Computer Science (EECS); Co-Director of Operations Research Center (ORC)

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ALEXANDRE JACQUILLAT, MIT

Assistant Professor of Operations Research (ORC)

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SAMORY KPOTUFE, Columbia University

Professor of Statistics

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